

Week 2 Office Hour Notes

2.0 Errata

9/9/24 6:00 PM. Added another Lemma between **Lemma 2.2.3** and what is now **Corollary 2.2.5**, and now **Corollary 2.2.5** has a second condition involving truth tables.

%DETAIL THE CHANGE THAT WAS DONE

9/10/24

2.1 Valuation, Truth Assignment Mappings, and Truth Tables

Definition 2.1.1. Let S be a set of sentence symbols (NOTE: it's just a set of SOME sentence symbols and NOT all of them)

1. A function $v : S \rightarrow \{T, F\}$ (some texts swap $\{T, F\}$ for $\{0, 1\}$) is called a **valuation mapping**
2. For every valuation mapping v , there exists an extension $\bar{v} : \bar{S} \rightarrow \{T, F\}$ called a **truth assignment mapping**, where $\bar{S}$ is the set of wff's that consist entirely of sentence symbols in S (think of $\bar{S}$ as the set of wff's "generated" by S) that satisfies the following rules:

For all wff's $\phi, \psi \in \bar{S}$, we have

1. $\bar{v}(\neg\phi) = \begin{cases} T & \text{if } \bar{v}(\phi) = F \\ F & \text{if } \bar{v}(\phi) = T \end{cases}$
2. $\bar{v}(\phi \wedge \psi) = \begin{cases} T & \text{if } \bar{v}(\phi) = \bar{v}(\psi) = T \\ F & \text{otherwise} \end{cases}$
3. $\bar{v}(\phi \vee \psi) = \begin{cases} T & \text{if either } \bar{v}(\phi) = T \text{ or } \bar{v}(\psi) = T \\ F & \text{otherwise} \end{cases}$
4. $\bar{v}(\phi \rightarrow \psi) = \begin{cases} T & \text{if either } \bar{v}(\phi) = F \text{ or } \bar{v}(\phi) = \bar{v}(\psi) = T \\ F & \bar{v}(\phi) = T \text{ but } \bar{v}(\psi) = F \end{cases}$
5. $\bar{v}(\phi \leftrightarrow \psi) = \begin{cases} T & \bar{v}(\phi) = \bar{v}(\psi) \\ F & \text{otherwise} \end{cases}$

Proposition 2.1.2. The extension $\bar{v}$ exists and is unique. (the codomain/range of $\bar{v}$ is uniquely determined; recall that in order for a function to be a function, the output value has to be a unique one)

Outline of Proof. I'll elaborate a little more the outline in an edit of these notes. Existence follows by using structural induction to show that we can assign unique and well-defined values as we go through (by structural induction) on the values that they would be (note that this is definition by recursion, which we talk about in week 4).

Uniqueness I'll leave to the reader, applying a straightforward structural induction argument.
□

Example 2.1.3. Let $S = \{A_1, A_2\}$ be a set of two (different) sentence symbols and define the valuation mapping

$$v : A_1 \mapsto T, A_2 \mapsto F$$

By **Proposition 2.1.2**, we have that the truth assignment $\bar{v}$ exists. We find that

$$\begin{aligned}\bar{v}(A_2) &= F, \\ \bar{v}(\neg A_1) &= F,\end{aligned}$$

Lastly, we find $\bar{v}(A_1 \rightarrow (A_1 \vee A_2))$ by first finding that

$$\bar{v}(A_1) = T, \bar{v}(A_2) = F, \bar{v}(A_1 \vee A_2) = T, \text{ so we conclude that}$$

$$\bar{v}(A_1 \rightarrow (A_1 \vee A_2)) = T.$$

In more informal iterations of sentential logic, we talked about truth assignment through what we call "truth tables", which more or less are a display of all possible valuations for valuation mappings $v : S_0 \rightarrow \{T, F\}$ such that S_0 is a finite collection of sentence symbols.

To see this, let's go back to **Example 2.1.3**, and explore the truth values for $A_1 \rightarrow (A_1 \vee A_2)$. Note here when we determine the value of the table by treating each row v_1, v_2, v_3, v_4 as its own valuation mapping, and all we're doing is applying the truth assignments $\bar{v}_1, \bar{v}_2, \bar{v}_3, \bar{v}_4$ each valuation row and getting the values using **Definition 2.1.1**.

A_1	A_2	$A_1 \vee A_2$	$A_1 \rightarrow (A_1 \vee A_2)$
T	T	T	T
T	F	T	T
F	T	T	T
F	F	F	T

We find that the formula $A_1 \rightarrow (A_1 \vee A_2)$ is true in all possible truth assignments for $S = \{A_1, A_2\}$, which tells us that it is a tautology (see next section!).

In light of this, we define truth tables in the following rigorous way:

Definition 2.1.4. Let τ be a wff, $\langle E_1, \dots, E_n, \tau_1, \dots, \tau_k \rangle$ the construction sequence of τ , starting with the distinct sentence symbols $E_1, \dots, E_n$, and $S_\tau := \{E_1, \dots, E_n\}$ be the set of sentence symbols in the construction sequence of τ . We define the **truth table generated by τ** to be the function $T_\tau : \{T, F\}^{S_\tau} \times \{E_1, \dots, E_n, \tau_1, \dots, \tau_k\} \rightarrow \{T, F\}$ defined by

$$T_\tau(v, \sigma) = \overline{v}(\sigma), \text{ where}$$

$$\{T, F\}^{S_\tau} := \left\{ v \subset \mathcal{P}(S_\tau \times \{T, F\}) : v \text{ is a valuation mapping } S_\tau \rightarrow \{T, F\} \right\}$$

and recall that $\overline{v} : \overline{S_\tau} \rightarrow \{T, F\}$ is the truth assignment generated by v .

Proposition 2.1.5. Let τ be a wff, $\langle E_1, \dots, E_n, \tau_1, \dots, \tau_k \rangle$ the construction sequence of τ , starting with the distinct sentence symbols $E_1, \dots, E_n$. The set $\{T, F\}^{S_\tau} \times \{E_1, \dots, E_n, \tau_1, \dots, \tau_k\}$ has 2^n rows and $n + k$ columns. Moreover, there is a 1-1 correspondence between the sequence of $\langle T_\tau(v, s_1), T_\tau(v, s_2), \dots, T_\tau(v, s_{n+k}) \rangle$ truth values of the rows of truth tables, for a fixed valuation mapping $v : S_\tau \rightarrow \{T, F\}$ and the set of truth assignments

$$\{T, F\}^{\overline{S_\tau}}.$$

%FIGURE OUT BETTER NOTATION FOR THIS

Proof.

%LEFT AS AN ESERCISIE

%TALK ABOUT PLANS FOR NEXT SECTION

2.2 Satisfiability and Tautological Implication

In general, let τ be a wff and Σ be an arbitrary set of wff's

Definition 2.2.1. We call Σ **satisfiable** if there exists a truth assignment mapping $\overline{v}$ such that $\overline{v}(\sigma) = T$, for every $\sigma \in \Sigma$, i.e. the entire set Σ is "true" in $\overline{v}$. We consider Σ **unsatisfiable** if no such truth assignment exists.

Definition 2.2.2.

1. We say that Σ **tautologically implies** τ (and write $\Sigma \models \tau$) if for every truth assignment $\bar{v}$ such that Σ is true in $\bar{v}$ (i.e. $\bar{v}$ satisfies Σ), we have that $\bar{v}(\tau) = T$. i.e., in all possible truth assignments where Σ is true, so is τ .

2. In the special case where $\Sigma = \emptyset$ (i.e., there are no formulas in Σ), we write $\models \tau$ whenever we have $\Sigma \models \tau$ in that scenario, and we call τ a **tautology**.

Lemma 2.2.3. We have $\models \tau$, i.e. τ is a tautology, if and only if for every truth assignment $\bar{v}$, we have $\bar{v}(\tau) = T$.

Outline of Proof. Immediate by the above definition (since Σ is the empty set, so there's no formulas we need to require true, so we quantify over all truth assignments in this case). $\square$

%EDIT PROOFS WITH THE RIGOROUS DEFINITION OF "TRUTH TABLES" IN MIND

Lemma 2.2.4. For every truth assignment $\bar{v} : \bar{S} \rightarrow \{T, F\}$ such that $\tau \in \bar{S}$, there exists a smallest finite set S_τ of sentence symbols such that the restriction of the valuation mapping $v|_{S_\tau} : S_\tau \rightarrow \{T, F\}$ $\tau \in \bar{S}_\tau$ and $\bar{v}|_{S_\tau}$ has a well-defined value. Moreover, the mapping $v|_{S_\tau}$ corresponds to some row in the truth table of τ

Proof. Let S_τ be the set of all sentence symbols that make up the construction sequence of τ , note that S_τ by construction is finite, $\tau \in \bar{S}_\tau$. we find that S_τ is the smallest such set since given $S' \subset S$ such that $\tau \in \bar{S}'$ must contain all sentence symbols that make up the construction sequence of τ (this is a fact that follows by straightforward structural induction on τ), hence we have $S_\tau \subset S'$.

The existence of a row of the truth table of τ that corresponds to $v|_{S_\tau}$ follows immediately from the fact that there is a 1-1 correspondence between all valuation mappings $u : S_\tau \rightarrow \{T, F\}$ and rows of a truth table. $\square$

Corollary 2.2.5.

1. τ is satisfiable if and only if its truth table has at least one row that yields the value T for τ
2. τ is a tautology if and only if every row of its truth table yields the value T for τ

Outline of Proof.

1. $\implies$ Suppose τ is satisfiable. We can choose $v : S \rightarrow \{T, F\}$ satisfying τ and by Lemma 2.2.4, we can choose a set $S_\tau \subset S$ such that the restriction $v|_{S_\tau}$ corresponds to a row of the truth table. Since $\overline{v|_{S_\tau}}(\tau) = \overline{v}(\tau) = T$, our conclusion that this row yields the value T .

2. $\implies$ follows immediately by **Lemma 2.2.3**, and $\impliedby$ follows from the fact that if we take an arbitrary truth assignment $\overline{v} : \overline{S} \rightarrow \{T, F\}$ generated by the valuation mapping $v : S \rightarrow \{T, F\}$, we can restrict that mapping to a row of the truth table on τ by looking at the finite set $S_0 := \{E_1, \dots, E_m\} \subset S$ containing all the sentence symbols that make up τ . $\square$

2.3 Questions

%add links to these questions

Question 2.3.1. What's the difference between the different principles of mathematical induction?

Answer: As mentioned before in the Week 1 office hour notes, there are various different *Principles of Mathematical Induction* involving the natural numbers $\mathbb{N} := \{1, 2, 3, \dots\}$, or certain "inductive subsets" of the integers $\mathbb{Z}$ (such as the nonnegative integers $\mathbb{N}_0 := \mathbb{N} \cup \{0\}$, or any subset of $\mathbb{Z}$ of the form $\mathbb{N}_m := \{n \in \mathbb{Z} : n \geq m\}$, for $m \in \mathbb{Z}$).

Typically, you have the two main such inductive principles:

1. *The Principle of (weak) Mathematical Induction*
2. *The Principle of Strong Induction*

and in addition, slide 24 of Dr. Pomper's Section 1.1 powerpoint (from week 1) goes over:

3. *The Principle of Structural Induction*

I explain each of these two different inductive principles in the aforementioned week 1 notes, but there are many different variations of these principles which you can find in other texts...including another way of wording the Principle of Mathematical Induction. During the week 2 discussion forum, I explained another way of defining the *Principle of Mathematical Induction* on Natural Numbers alongside the *Principle of Structural Induction*



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Here are the two different principles of induction (in one of their possible phrasings) side-by-side:

Principle of Mathematical Induction: Let $S \subset \mathbb{N} := \{1, 2, 3, \dots\}$. If S has the following conditions:

1. (atomic condition) $1 \in S$,
2. (inductive step) If $n \in S$, then $n + 1 \in S$.

Then $S = \mathbb{N}$.

Principle of Structural Induction (on wff's): Let S be a collection of wff's. If S has the following conditions:

1. (atomic condition) $A_n \in S$, for every sentence symbol A_n ,
2. (inductive step) If $\alpha_1, \alpha_2 \in S$, then $\neg\alpha_1, \alpha_1 \wedge \alpha_2, \alpha_1 \vee \alpha_2, \alpha_1 \rightarrow \alpha_2, \alpha_1 \leftrightarrow \alpha_2 \in S$,

then S consists ALL wff's.

Hopefully you can see the parallels of the two, but if that doesn't answer all of your curiosity on the matter, we will be tying all the concepts of mathematical induction together in week 4.

△

Question 2. What's a certain notation symbol (which we'll see later)